

Measuring the Differential Cross Section of Boosted Higgs Bosons in the $H \rightarrow WW^* \rightarrow qq\ell\nu$ Channel as a Probe for BSM Physics

Peilin Ye, Professor Dylan Rankin
University of Pennsylvania

September 8, 2025

Objective and Significance

The objective of this project is to measure the differential cross section of boosted Higgs boson decays to WW^* in the semi-leptonic channel ($H \rightarrow WW^* \rightarrow qq\ell\nu$). By focusing on the boosted (high transverse momentum (p_T)) regime, the goal is to enhance sensitivity to heavy beyond-the-Standard-Model (BSM) particles that couple to the Higgs. Since the Higgs couples to all massive particles, any heavy BSM particle can leave indirect imprints on Higgs production rates. These effects would create larger deviations at high p_T due to their heavy mass. Consequently, precise differential cross section measurements provide a powerful indirect probe for new physics complementary to direct searches.

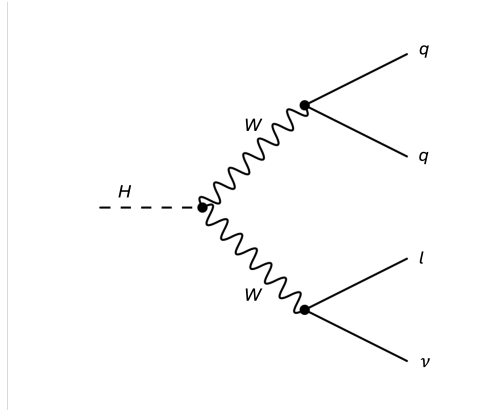


Figure 1: Feynman diagram of $H \rightarrow WW^* \rightarrow qq\ell\nu$

Background and Preliminary Results

The Higgs boson, discovered in 2012 by ATLAS and CMS [1,2], is central to electroweak symmetry breaking. Its couplings to fermions and vector bosons have been measured with increasing precision, largely in agreement with Standard Model (SM) predictions [3]. However, the possibility remains that the Higgs is a portal to heavy BSM states beyond the direct kinematic reach of the LHC.

One promising avenue is to study differential distributions of Higgs production, particularly the transverse momentum spectrum. In processes dominated by gluon-gluon fusion (ggF) and vector

boson fusion (VBF), loop-level contributions from heavy particles may alter the high- p_T tail of the Higgs cross-section. Thus, any deviation from SM predictions could indicate new physics.

The $H \rightarrow WW^*$ decay mode is especially interesting due to its relatively large branching ratio ($\sim 21\%$) and W -boson being charged. We focus first on the semi-leptonic final state ($qql\nu$), which balances manageable backgrounds with full reconstruction of Higgs kinematics. Considering the branching ratio of $W \rightarrow qq$ ($\sim 68\%$) and $W \rightarrow \ell\nu$ ($\sim 32\%$), this final state is a good balance for the overall event yield ($\sim 44\%$ of all WW final states).

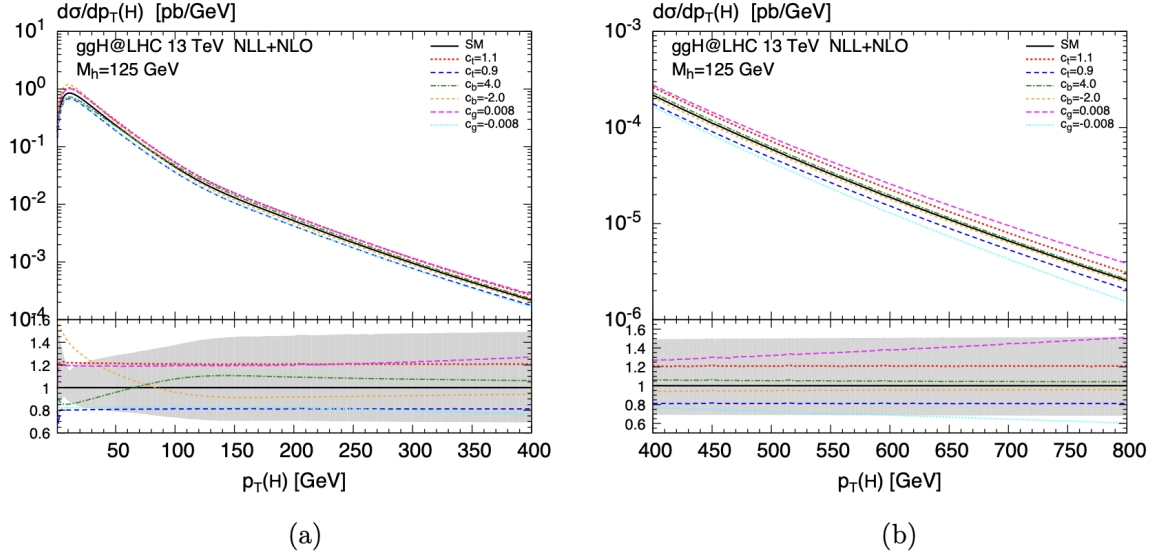


Figure 2: Expected cross-section with/without BSM contribution affecting Higgs coupling strength. [4] The black line is the SM prediction, while the other dashed lines are the prediction after some BSM mechanism affecting Higgs coupling strength.

Some studies from ATLAS and CMS are considering BSM effects on the Higgs cross-section spectrum [4] but their sensitivity can be improved. Early results suggest ML models can improve signal sensitivity compared to traditional cut-based methods, and further improvements on ML models can help.

Work Plan

The project will proceed along the following directions:

Event Selection and Background

We will target Higgs events produced primarily via ggF and focus on events with large boosts in the transverse plane. We will select events with one lepton (electron or muon), large missing transverse energy (E_T^{miss}), and (ideally) two hadronic jets. We are particularly interested in the boosted regime, where the two hadronic jets will be close enough to be identified as a wide (large-R) jet. We will study the jet substructure to identify if it is from a hadronic W decay. For this decay channel, the main background is $W + \text{jets}$.

Machine Learning Classifiers

The main focus of this project is studying different ML architectures and their performance. We will develop ML-based discriminants to separate signal from background rather than traditional cut-based methods, since boosting can make particle candidates cluster and traditional methods work poorly. Candidate models include:

- Dense Neural Networks (DNNs) as the baseline model.
- Graph Neural Networks (GNNs) to exploit jet substructure topology.
- Attention-based models to capture correlations across final-state objects.

Extensions

Depending on feasibility, we may extend the analysis to other final states ($\ell\ell\nu\nu$, $qqqq$) to cross-check consistency. We may also explore model calibration and background normalization if time permits.

Cited References

References

- [1] ATLAS Collaboration, *Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC*, Phys. Lett. B716 (2012).
- [2] CMS Collaboration, *Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC*, Phys. Lett. B716 (2012).
- [3] D. de Florian et al., *Handbook of LHC Higgs Cross Sections: 4. Deciphering the Nature of the Higgs Sector*, CERN Yellow Reports, CERN-2017-002-M.
- [4] M. Grazzini et al., *Modeling BSM effects on the Higgs transverse-momentum spectrum in an EFT approach*, JHEP (2017).